

Response Letter with Annotations

Lu, L., Kwon, H., Wang, W., M. Cotter, L., A. Kriss, L., Minich, M., Liu, J., N. Cascio, C., & Yang, S. (2025). Cannabis warning labels, sensory marketing, and electronic word-of-mouth: AI-facilitated textual analysis of a randomized experiment among youth and young adults. *International Journal of Advertising*, 1–26. <https://doi.org/10.1080/02650487.2025.2548647>

Annotations

1. Purposes

a. Motivation (one-sentence summary)

The research examines effects of cannabis warning labels (CWLs) and social media comments in advertising on eWOM.

b. Specific Rationale (several bullet points)

- CWLs uses sensory marketing that appeals to youth.
- There is limited research on studying communication-related factors such as word-of-mouth.
- Authors apply the MOA framework and predict CWLs and comments should predict WOM

c. Research Questions and/or Hypotheses

• Identify IV, DV, mediator, moderator

- IV: Cannabis warning labels (CWLs), social media comments
- DV: Stance of youth WOM (pro-cannabis and anti-cannabis), valence of youth WOM, sensory marketing appeals
- Moderator: social media comments

• Identify specific relationships among variables

- CWLs will (a) increase anti-cannabis and (b) decrease pro-cannabis youth WOM.
- (a) Exposure to pro-cannabis comments will increase pro-cannabis word-of-mouth, whereas (b) seeing anti-cannabis comments will increase anti-cannabis word-of-mouth toward cannabis marketing posts.
- Author propose some RQs without specifying directions (e.g., moderation effect, effects on sensory marketing appeals).
- Authors also propose a RQ about themes.

2. Method

a. Research Design

- Experiment: 3 (warning labels: no warning vs. textual warning vs. pictorial warning) $\times$ 3 (social media comments: no comment vs. pro-cannabis comments vs. anti-cannabis comments) between-subjects factorial design

b. Sample

- 3097 participants between 13 and 25 from the Qualtrics
- Not representative but fits with their research goal

c. Measures

- IVs are manipulated
- Structural topic modeling is used for identifying themes.
- Stance of youth WOM is labeled by GPT.
- Valence is measured by multiple procedures: (1) structural topic modeling, (2) manual coding, (3) GPT classification.
- Sensory appeals are manually coded first and then GPT classified.

d. Analysis

- Overall analysis could be summarized into (1) inductive analysis for STM, (2) deductive analysis for GPT classification, (3) logistic regression.

3. Results and Conclusions

a. Answers to RQs and Hypotheses

- CWLs significantly increases anti-cannabis and decreases pro-cannabis youth WOM (supported).
- (a) Exposure to pro-cannabis comments does not pro-cannabis word-of-mouth (not supported), whereas (b) seeing anti-cannabis comments significantly increase anti-cannabis word-of-mouth toward cannabis marketing posts (supported).
- Comments do not function as a moderator.
- Authors identify eight themes in each stance WOM.
- CWLs and comments significantly reduce sensory appeals.

b. Include Key Figures/Tables

Table 1. Inductive analyses through structural topic Modeling.

Topic	Percentage	Top words	Examples
T2 Anti-Cannabis & Intervention Intention	40%	don't, want, not, stop	I don't want anything to do with drugs and this needs to stop. Do not do drugs for fun.
T1 Positive Experience	25%	edible, yummy, delicious	MmmmMmmm!!!! Looks yummy. I would never think that that was an edible!!! Looks very delicious! Hope it taste as good as it looks!
T3 Urge & Attraction	20%	appealing, perfect, good	I think It's appealing and good product. They're extremely good and perfect for you and me.
T4 Perceived Risk	15%	bad, effect, side	Marijuana is a harm drug it's not to be taken lightly, it's bad and can cause serious damage. A lot of people aren't aware of the possible side effects.

Table 2. GPT prompts with few-shot learning for stance classification.

Category	GPT prompt
Pro-Cannabis Stance	What is the stance of the text below with respect to 'cannabis'? If we can infer from the text that the text expresses a positive or supportive stance on cannabis, please label it as 'positive'. If we can infer from the text that the text does not express any positive or supportive stance on cannabis, meaning the text does not fall into positive, please label it as 'non-positive'. Please use exactly one word from the following two categories to label it: 'positive' 'non-positive'. Here are some examples of texts. Makes sure to classify the last text correctly. Q: text: this looks interesting. looks cool might want to try this A: positive Q: text: marijuana helps me ease my tension and i love it cannabis makes people feel relaxed and happy, and makes new friends marijuana can lift my mood and make me more productive A: positive Q: text: looks very delicious! hope it taste as good as it looks! wow! i have never seen edibles like this! where i live i've never had the option for a candy bar. does it come in milk chocolate? A: positive Q: text: this is not good nor is it healthy for people! stop this madness! this is harming people!! this is horrible A: non-positive Q: text: ok ok ok A: non-positive Q: text: thanks for helping my education im gon not use weed no more weed A: non-positive Q: text: '{text}' Is this text positive or non-positive? A:
Anti-Cannabis Stance	What is the stance of the text below with respect to 'cannabis'? If we can infer from the text that the text expresses a negative or critical stance on cannabis, please label it as 'negative'. If we can infer from the text that the text does not express any negative or critical stance on cannabis, meaning the text does not fall into negative, please label it as 'non-negative'. Please use exactly one word from the following two categories to label it: 'negative' 'non-negative'. Here are some examples of texts. Makes sure to classify the last text correctly. Q: text: i enjoy feeling at ease myself i do not want to drink thc nothing comes to mind A: non-negative Q: text: marijuana helps me ease my tension and i love it cannabis makes people feel relaxed and happy, and makes new friends marijuana can lift my mood and make me more productive A: non-negative Q: text: can it be used by teenagers and does it have significant harm? did you prove that warning? A: non-negative Q: text: looks very delicious! hope it taste as good as it looks! wow! i have never seen edibles like this! where i live i've never had the option for a candy bar. does it come in milk chocolate? A: non-negative Q: text: this is not good nor is it healthy for people! stop this madness! this is harming people!! this is horrible A: negative Q: text: can't drive after smoking marijuana or it will cause a car accident smoking too much can affect your health long-term smoking affects mental problems A: negative Q: text: thanks for helping my education im gon not use weed no more weed A: negative Q: text: that's bad for you edibles are bad for you. marijuana is a harm drug it's not to be taken lightly, it's bad and can cause serious damage. A: negative Q: text: '{text}' Is this text negative or non-negative? A:

Table 3. Effects of CWLs and prior eWOM on stance towards cannabis.

	Pro-cannabis stance		Anti-cannabis stance	
	Main effect	Interaction	Main effect	Interaction
Intercept	0.40*** (0.227)	0.45*** (0.240)	0.26*** (0.346)	0.21*** (0.392)
Pictorial warning	0.54*** (0.107)	0.44*** (0.189)	1.68** (0.169)	2.65** (0.316)
Textual warning	0.58*** (0.108)	0.45*** (0.195)	1.67** (0.172)	1.79 (0.349)
Pro-cannabis comment	0.92 (0.109)	0.73 (0.172)	1.00 (0.177)	1.25 (0.333)
Anti-cannabis comment	0.74*** (0.111)	0.63*** (0.178)	1.50* (0.169)	2.12* (0.316)
Pictorial warning X Pro-cannabis comment		1.57 (0.261)		0.60 (0.437)
Textual warning X Pro-cannabis comment		1.38 (0.263)		0.97 (0.458)
Pictorial warning X Anti-cannabis comment		1.20 (0.266)		0.46 (0.417)
Textual warning X Anti-cannabis comment		1.44 (0.271)		0.87 (0.438)

Note: The values shown are Odds Ratios (OR), with standard errors in parentheses. The logistic regression analysis controlled for 7 covariates such as race, gender, income, sexual orientation, Hispanic, political ideology, and social environment ($N=2229$). * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 4. Thematic analysis for pro- and anti-cannabis comments.

Stance	Theme	Example
Pro-cannabis	Sensory Appeals	how interesting an innovative i like that it is a candy that will mask the flavor of the marijuana definitely a product that i would like to try looks good would love to try oh id love to try this because of the popcorn flavor. omg i love snickerdoodles we need more details of the product lol it looks like candy so i lowkey want to try i love chocolate so i might check it out!!!
	Health Benefits of Cannabis	marijuana, helps with depression adhd, and many different beneficial factors. if mary jane aka marijuana was introduced and cigarettes weren't or any tobacco products there would be alot more healthier births, less drama or violence. plus stores would be helped out more with sales do to the thc factor in marijuana. i believe that cigarettes should be banned and to leave mary jane blessing her people. nothing but pros with this topic, marijuana helps with a lot of health problems and mental issues. i've never heard anyone i mean nobody having negative effects where they needed medical attention. for the ones that talk bad on it they never tried it and there information is just from other uneducated people that have never tried or did research. if you have the time look up deaths from marijuana and that will give your answer on how safe it is to consume or smoke. stay woke and with your mind in the horizons, marijuana never hurt people but instead did good inlike cigarettes and alcohol that kills your liver teeth and heart. "i think cannabis is medicine. stop calling it marijuana i just think you guys seem bias to new information about cannabis benefits can't make cannabis sound bad to me" i love me some good ol' mary jane. good for anxiety, manic depression, and in my case, schizophrenia. it really really helps my paranoia, but i know that everyone is different. i love the idea. the me, the flavor. all of it looks great. it just looked a little bit boring. i was confused, i thought i was looking at thc infused hand soap.
	Recreational Benefits of Cannabis	I love weed high sex is the best sex high sex is the best sex weed is so good i'd love to have some of this rn having marijuana edibles rn would make me feel on top of the world weed and wine would definitely go hand in hand I love the loud smoking keeps me going I like smoking cause it gives me knowledge
	Support for Legalization/Regulation	marijuana like alcohol or caffeine can be abused. legalizing marijuana is not going to increase abuse however, it will help people that are disproportionately effected by policing to be criminalized. marijuana is as addictive as caffeine and alcohol. the effects one person might have from consuming marijuana is different to another person. caffeine can cause hypertension and anxiety. alcohol dependence can cause cirrosis of the liver. i wonder why only bad comments are being shown. also, the facebook layout makes me think about the demographic they are advertising to. i wish it was legal here if only it was legal in ky warning lable is false! glad to see this is becoming regulated .! glad to see regulation !!! whoooo whoop great to see this on this market regulated ! whoooo hoo
Anti-cannabis	Safety and Health Concern	smoking weed is bad for your brain. looks yummy but is it safe weed will mess you up this is not good nor is it healthy for people! stop this madness! this is harming people!! this is horrible Marijuana is bad for people; it may cause health issues! I do not like anything about marijuana; I hate edibles.
	Marketing Critique	I don't like the marketing and possibly targeting the wrong crowd with the 'appealing' look. Shameful marketing, and sad to see no one else in this comment section sees or understands this. "i'm not a fan of the marketing and possibly targeting the wrong crowd"
	Concerns about Youth Consumption	Marijuana should be targeted towards medical users like me, instead of this kiddish style marketing another "groundbreaking" marijuana product brainwashing kids with those bright colors trendy brands trying to make weed cool! i would not use this at all. its stupid its a bad product. and stupid to show a kid. its not a product for kids
	Personal Narratives as Intervention	I have seen the effects happen to people in my own life, and it's not good. I started smoking really young (7th grade) and have an awful memory; sometimes i wonder if my early cannabis usage is the reason for that. My little brother suffers from schizophrenia that was brought on by his first time smoking weed.

Table 5. Descriptive statistics about specific themes by stance.

Stance	Theme	Estimate count (%)
Pro-cannabis	Favorable opinion toward the cannabis product shown in the marijuana marketing post	140 (70%)
	Health benefits of cannabis in general	5 (2.5%)
	Recreational benefits of cannabis in general	11 (5.5%)
	Support for legalizing/regulating marijuana	7 (3.5%)
Anti-cannabis	Safety concern and health risk	40 (20%)
	Marketing strategy critique	4 (2%)
	Concerns about Youth/Underage consumption	17 (8.5%)
	Use personal narratives for intervention	10 (5%)

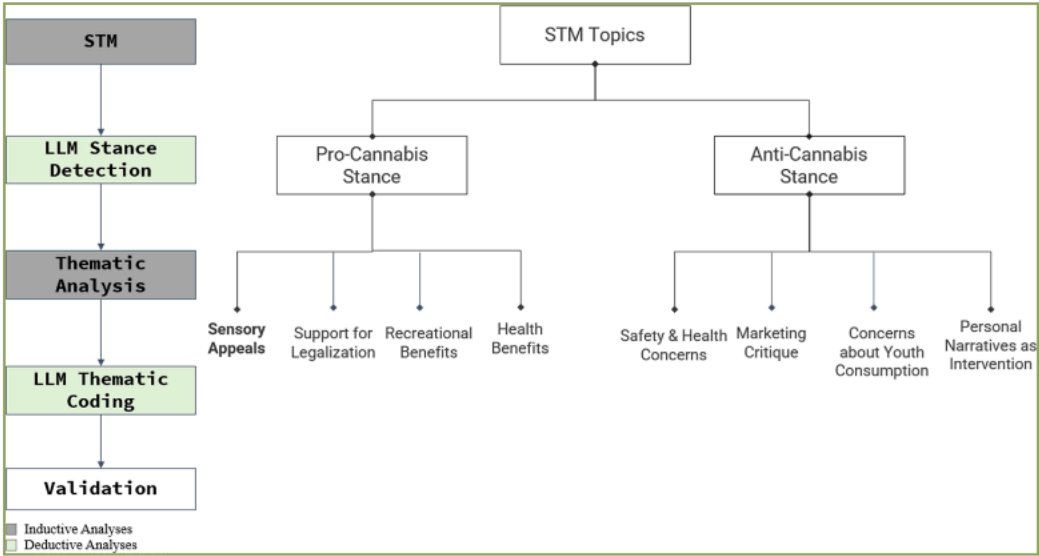


Figure 1. AI-facilitated integrated analysis flowchart.

Table 6. Effects of CWLs and prior eWOM on mention of sensory appeals.

	Main effect	Interaction
Intercept	0.33*** (0.288)	0.35*** (0.295)
Pictorial warning	0.43*** (0.141)	0.36*** (0.230)
Textual warning	0.53*** (0.136)	0.45*** (0.226)
Pro-cannabis comment	0.75* (0.132)	0.63* (0.188)
Anti-cannabis comment	0.53*** (0.144)	0.49*** (0.203)
Pictorial warning X Pro-cannabis comment		1.54 (0.326)
Textual warning X Pro-cannabis comment		1.32 (0.319)
Pictorial warning X Anti-cannabis comment		1.04 (0.363)
Textual warning X Anti-cannabis comment		1.24 (0.347)

Note: The values shown are OR, with standard errors in parentheses. For efficient interpretation, it is helpful to transform the OR to a percentage. The logistic regression analysis controlled for 7 covariates such as race, gender, income, sexual orientation, Hispanic, political ideology, and social environment ($N=2229$). * $p < .05$, *** $p < .001$.

c. Authors' Conclusions

Cannabis warning labels (CWLs) could shape interpersonal communication among youth. Anti-cannabis comments can amplify negative WOM, but pro-cannabis does not work. Youth may resist to promotional content in this context.

4. Discussion

a. Implications

- Regulatory entities can strategically deployed warning labels to reduce emotional appeals among youth.
- Advertisers should be cautious of using pro-product eWOM to avoid any backfire.
- The research using AI-based textual analysis in mixed-methods experimental design is methodologically innovated.
- Results support the MOA framework and limited capacity model of motivated mediated message processing.

b. Limitations

- Product scope is limited to certain cannabis products.
- Online experiment not in a real setting.
- Intercoder reliability for anti-cannabis comments is lower than 0.80.

c. Future Directions

- Expand to other cannabis products.
- Directly measure and test the mediating mechanisms of positive and negative emotions and knowledge gain.

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Response Letter (~600 words)

Lu et al. (2025) want to test how cannabis warning labels (CWLs) and social media comments influence the electronic word-of-mouth (eWOM) that adolescents and young adults produce in response to online cannabis advertising. Overall, I think this is an excellent research, based on the authors' research goal, theoretical foundation, and method innovation. The most important thing is that this article shows how AI could participate in coding procedure instead of content analysis by human coders.

In terms of theory, I think the theoretical foundation is strong, appropriate, and well-justified for the authors' research questions. Given authors' exploration on related literature, this project examines effects of cannabis warning labels (CWLs) in advertising on eWOM to address a gap in this literature. This research is grounded on multiple theoretical frameworks, including the motivation, opportunity, and ability theoretical (MOA) framework, the limited capacity model of mediated message processing, and social cognition theory to justify rationales for their main variables and relationships. However, some elements in the MOA framework are not fully operationalized. As the authors acknowledge that they should further explore knowledge or ability as a mediator in the future.

Regarding their method, the authors employed mixed method approach, which is reasonable and innovated. This also reminds of the instructor talking about incorporating computational method in an online experiment, and I think this article is a good example. In designing experiments, I appreciate the authors' effort in design stimuli following a validated protocol. The experiment seems to have too many participants, and I'm not sure this is intentionally done because they need a large amount of data for structural topic modeling. As this week's topic is about LLM, I think the most important part in this article is about their pipeline, which starts with an inductive approach to see what is happening in youth's comments, and uses deductive approach for classification and causal inference. We have learned the structural topic modeling in previous weeks, and this section is very clear. For classification, the authors first develop a benchmark dataset based on human coding around 10% and then use GPT as a core analytic tool to classify thousands of open-ended comments generated by participants as supervised machine learning. The authors also keep the prompt and the procedure very transparent. The performance evaluation is also good on three aspects, despite one category has a lower reliability. For their analysis, they use logistic regression for binary coding outcomes from previous steps. Personally, I really like this research's pipeline in their method, which is very inspiring for other future research.

Overall, I think the method sections is very impressive, and strengths definitely outweigh its limitations. For some limitations, the authors already address several limitations. For instance, the product scope is limited to certain cannabis products, and intercoder reliability for anti-cannabis comments is lower than 0.80. Besides these, I think the current manipulation of eWOM is little hypothetical. Current research can be improved by doing this experiment in a real setting, such as collaborating with an advertising campaign. Another option is to conduct a quasi-experiment based on two different existing advertising campaigns and collect online users' comments.